Model and Data Card: Illegal Brick Kiln Detector

Fine-tuned YOLO11n-OBB for satellite brick kiln detection in Bangladesh review workflows

Team PTSD | AI Hackathon SciBlitz AI Challenge 2026
Samonwita Sarker (Leader), Protik Dev, Rotna Dipika Debnath, Touhidul Alam Seyam

Model details

Field	Value
Model	YOLO11n-OBB, pretrained `yolo11n-obb.pt`
Provider	Ultralytics pretrained checkpoint, fine-tuned in project notebook
Artifacts	`model/best.pt` (5.5 MB), `model/best.tflite` (10 MB)
Task	Oriented object detection for brick kiln review
Classes	CFCBK, FCBK, Zigzag
License	Verify dataset and model licenses from the original providers before public redistribution

Training data

Field	Value
Dataset	SentinelKilnDB from Hugging Face, SustainabilityLabIITGN/SentinelKilnDB
Format	Parquet source converted to PNG tiles plus YOLO-OBB labels
Split used	Full South Asia train/val; Bangladesh-only validation subset from val
Counts	71,856 train tiles, 23,952 val tiles, 3,929 Bangladesh val tiles
Class counts	Train labels: CFCBK 2,032; FCBK 34,292; Zigzag 27,463

Training setup

Field	Value
Notebook	`yolo-obb-n (1).ipynb`
Image size	128 px in completed notebook run
Epochs/batch	50 epochs, batch 256
Augmentations	Rotation, horizontal flip, vertical flip, scale, translate, mosaic
Export	LiteRT via `scripts/export-litert.py`; input `(1,3,256,256)`, output `(1,8,1344)`

Validation results

Class	P	R	mAP50	mAP50-95
All	0.651	0.627	0.643	0.359
CFCBK	0.661	0.572	0.592	0.374
FCBK	0.618	0.672	0.634	0.330
Zigzag	0.675	0.636	0.702	0.375

Intended use

- Triage likely brick kiln locations from satellite crops.
- Review map markers, confidence, tile evidence, and OBB overlays.
- Support environmental monitoring. It is not a standalone enforcement system.

Known limitations

- Saved outputs include overall validation metrics, not final Bangladesh-only metrics.
- Demo tile coverage is limited to five districts and ten seeded tiles per district.
- Upload predictions without georeferenced tiles have approximate coordinates.
- Class imbalance is substantial; FCBK dominates the train labels.
- Visual false positives can occur for industrial roofs, roads, fields, and kiln-like shapes.

Ethical and deployment concerns

- Human review is required before enforcement, public accusation, or site-level publication.
- Satellite imagery and model outputs can be outdated; field validation should confirm sensitive findings.
- Thresholds should match the review context: high recall for screening, higher precision for action.
- Dataset and pretrained model licenses must be checked before external release or commercial reuse.